

SPECIALIZED PROJECT (ĐACN) · HCMUT

JobQuest

An AI-native experiential career-exploration platform

"Try the job before you pick the career."

Report on Chapters 1-5 · 18 September 2026

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Twenty minutes, five parts

Three of the five parts carry a decision we need from you today.

1 Problem and mandate

Chapter 1

2 Evidence base

Chapter 2

3 Related work and the gap

Chapter 3 · ASK 1

4 The proposed system

Chapter 4 · ASK 2

5 Analysis, evaluation, next steps

Chapter 5 · ASK 3

This deck reports Chapters 1 to 5.1. Implementation, testing, evaluation and conclusions follow in Chapters 6 to 8 (§1.5).

The decision is made blind

People choose an occupation without knowing what the work is.

1

The obstacle is informational

In Gati, Krausz and Osipow's validated taxonomy, insufficient occupational information is a principal category of career decision difficulty — distinct from indecisiveness and from poor self-knowledge. [1]

2

What guidance offers instead

Interest and personality inventories elicit what a person believes about themselves. They do not supply information about the work, so they cannot address that category of difficulty.

39–76%

of respondents are assigned a different MBTI type on retest after five weeks [5]

Two implications: the deficit that most needs fixing is informational, and the dominant instruments are structurally unable to fix it.

[1] Gati, Krausz & Osipow (1996), cross-validated across 13 countries. [5] MBTI test–retest, five-week interval.

The Vietnamese case

Compressed to age 17 or 18 — and often not the student's own decision.

> 1 in 5

chose their field on parental direction

~ 1 in 5

entered with no defined occupational goal

~ 1 in 3

were later unsure they would choose the same field again

The downstream cost. Analysis of successive Labour Force Survey waves finds the share of graduates in fully matched employment declining, with vertical and combined mismatch carrying measurable wage penalties. [4]

Honest previews of a job improve fit

Established, not speculative — four decades of meta-analytic evidence.

1

Phillips, 1998 — the effect

Meta-analysis of 40 studies. Realistic job previews reduce inflated expectations, increase self-selection, improve performance and lower voluntary turnover. Richer audiovisual previews outperform written ones. [6]

2

Earnest, Allen & Landis, 2011 — the mechanism

Meta-analytic path analysis over 52 studies and approximately 17,000 participants identifies perceived honesty as the mediating mechanism. The effect depends on candour, not on information volume. [7]

The constraint was never evidence — it was delivery. Previews required workplace exposure, or content authored one occupation at a time.

[6] Phillips (1998), 40 studies. [7] Earnest, Allen & Landis (2011), 52 studies, ~17,000 participants.

Objectives, scope and non-goals

Scope is restricted for method, not convenience — that makes claims testable.

Five objectives

- 1 Constrain generation so scenarios stay occupationally faithful.
- 2 Establish whether automated assessment agrees with experts sufficiently for formative use.
- 3 Design competence-based progression across occupation5.
- 4 Determine which game mechanisms sustain engagement without displacing learning.
- 5 Construct an offline evaluation methodology capable of substantiating or refuting the above.

Scope

Phase 1 covers the information technology field — complete in function, bounded in coverage. The design is domain-agnostic even though the evaluated instance is not.

Explicitly not

- Longitudinal career outcomes
- Recruitment or application brokering
- Any claim that simulation substitutes for employment

Why this is worth doing — practically and scientifically

Two kinds of significance, justifying two different things.

Practical

A means of acquiring occupational information experientially, before commitment — for students choosing a field, workers considering transition, and guidance provision generally. It rests on three established findings:

- Balanced, honest previews improve fit and reduce later regret [6][7]
- The obstacle they address is a principal source of decision difficulty [1]
- Successful performance of authentic tasks raises task-specific self-efficacy, which social cognitive career theory identifies as a determinant of choice and persistence [8][9]

For Vietnam specifically: an alternative to a guidance market composed almost entirely of self-report instruments.

Scientific

The contribution is combining and validating techniques demonstrated separately but not together:

- Constrained generation of domain-accurate scenarios
- Unobtrusive assessment of competence from behaviour
- Automated judgment of open responses

Each carries a documented weakness. The methodological contribution is an evaluation design that measures those weaknesses directly rather than assuming them resolved.

Doing beats reading

Doing a task conveys what description cannot — far beyond clinical training.

1

Kolb — the mechanism

Learning transforms concrete experience, through reflection, into conceptualisation. An occupational description supplies the concept without the experience the model says understanding is constructed from. [10]

2

Chernikova et al., 2020 — the evidence

Meta-analysis of 145 studies in higher education. Large overall effect that generalises across academic domains, with scaffolding as a moderator: worked examples for novices, reflective prompts for advanced learners. [16]

Design consequence. Scaffolding differs by learner level — which is why novice scenarios in JobQuest carry more context than advanced ones.

Measuring competence from behaviour

Performance replaces self-report — evidence-centered design specifies how.

Model	What it does	The requirement it imposes
Competency model	Defines what is claimed to be measured	Constructs must be specified independently of any one task if inferences are to transfer
Evidence model	Maps behaviour to inferences	Every observable must be attributable to a named construct
Task model	Specifies situations eliciting the behaviour	The construct must be NECESSARY to completing the task, not merely permitted by it

The third requirement is the most demanding and the most frequently unmet.

If a task can be completed by a route that bypasses the target construct, the evidence model is measuring something other than what it claims. Authoring a task is therefore a psychometric activity, not an editorial one — which is why UC-08 gives the domain expert an explicit exception path when a scenario turns out to be passable without the target competency.

Career theory: fit is not a fixed verdict

Four traditions — and two of them say something most career systems ignore.

1

Holland

Occupational choice as person–environment congruence. Underlies the interest inventories in widespread use, including the public-domain instrument distributed with O*NET. [32]

2

Social cognitive career theory

Self-efficacy, built principally through mastery experience, determines interest formation and choice — accessible to a system where users perform tasks, inaccessible to one that only elicits self-description. [8][9]

3

Super

Vocational self-concept develops over time, so fit is changing rather than fixed. [33]

4

Krumboltz

Unplanned events are a normal and potentially productive part of career development, not disruption. [34]

Consequence. Super and Krumboltz jointly justify two features: probabilistic life events (FR-14), and treating work–life balance as an attribute of an occupation rather than an externality (the five-attribute rating, FR-12).

Two findings that bound what we may claim

Both cut against the claim a project like ours usually makes.

Gamification does little for competence

0.638

g = autonomy

1.776

g = relatedness

0.277

g = competence

Hedges' g by psychological need [24]. Competence is the outlier.

Consequence: skill development must rest on task authenticity and validated assessment — never on reward structures.

Simulation works as a supplement, not a substitute

Across 65 studies, simulation-taught trainees exceeded comparison groups on self-efficacy, declarative and procedural knowledge and retention — but only where the game actively conveyed material, permitted learner control, and supplemented other instruction. As a sole method, comparison groups did better. The analysis also reported evidence of publication bias. [27]

Consequence: JobQuest informs occupational choice and builds initial self-efficacy. It does not substitute for employment or for formal instruction.

Generation and judging

LLMs can judge open responses — but judging must be validated in four ways.

1 • Feasibility

Agreement with human preferences above 80%, comparable to inter-human agreement — while the same work documents position, verbosity and self-preference bias. [44]

2 • Method

Structured chain-of-thought against an explicit rubric improves alignment, with a residual preference for model-generated text. [45]

3 • Validation

Raw agreement systematically overstates chance-corrected performance: ~41-point discrepancy between exact-match agreement and Cohen's κ on one benchmark [46]. On deliberately difficult pairs with objective labels, the strongest general-purpose evaluator reached ~64% accuracy [47].

The four requirements

- ✓ Report agreement with chance correction
→ *evaluation design*
- ✓ Draw the evaluator from a different model family than the generator
→ *BR-09*
- ✓ Control option ordering and response length
→ *assessment pipeline*
- ✓ Report performance stratified by item difficulty
→ *reference-set strata*

What we build on: taxonomies and benchmarks

We ground claims in established resources rather than collecting bespoke data.

Occupational taxonomies

O*NET

~900 occupations

linked to more than 19,000 task statements across some 2,000 detailed work activities [49]

ESCO

3,039 occupations

and 13,939 skills across 28 languages, with an official correspondence to O*NET [50]

Both express occupations as compositions of skills and activities — which is what supplies an external basis for occupational adjacency (BR-03, DR-01–DR-03) and a source for authoring representative tasks.

Evaluation benchmarks

Automated assessment

MT-Bench and the arena methodology [44] · JudgeBench [47] · RewardBench [51]

Generated instructional content

EQGBench [52] · EduBench [53]

Agents in branching narrative

TextWorld [54] · Jericho [55]

These give us a reporting methodology, not only reference data. No bespoke data collection is required for either grounding or evaluation.

[49] O*NET. [50] ESCO. [44][47][51] judging benchmarks. [52][53] generated instructional content. [54][55] agents in branching narrative.

Research prototypes

The direction is active and credible. Three limitations recur.

1

Reflection is the terminal outcome

CareerSim advances an avatar through life stages to produce a report; 12 participants; interaction concerns a trajectory, not the tasks constituting an occupation [57]. Future You reduced anxiety and raised future self-continuity across 344 participants [58]; Letters from Future Self found comparable benefits with 36 [59]. Neither lets a user discover through performance that the imagined self would dislike the work.

2

Trajectory abstraction without task fidelity

CareerPooler reports significant gains in engagement, information gain, satisfaction and career clarity over a conversational baseline across 24 participants — the strongest available evidence that generated interactive exploration beats conversational advice. But the representation is metaphorical, so outcomes are clarity, not demonstrated capability. [60]

3

Guidance without measurement

An SDT-grounded counselling agent reduced decision difficulty — measured by self-report. [63]

The binding limitation is evaluation practice: sample sizes 12–36, one at 344; designs predominantly qualitative or within-subject. Right for feasibility, insufficient for a claim about assessment accuracy.

Deployed systems

Each solves part of the problem. None solves the combination.

International

1

Forage

Employer-authored simulations with real work products. Proves users will voluntarily attempt occupational tasks. But content only expands as employers commission it, and assessment ends in a fixed model answer, not the user's own response. [64]

2

EntryLevel / Anthropos

Generated simulations and multi-role workplace scenarios producing behavioural evidence. Anthropos is the closest deployed analogue to behavioural assessment — but oriented to personnel selection, so the evidence serves an employer's hiring decision rather than a user's exploration. [65][66]

3

Hidden Door

Generates narrative within bounded authored components, with the reported consequence that narratives stay coherent across sessions. Commercial corroboration that authored structure, not model scale, governs consistency. [70]

Vietnam

**JobWay · JobTest.vn · Hướng nghiệp
Sông An · Hướng Nghiệp LwL · AI
Hay**

Assessment, occupational content and counselling, with recent entrants adding conversational AI. [71–75]

**No surveyed Vietnamese system
offers generated, performed
occupational simulation.**

How the criteria were derived

Comparing systems against our own features would be circular.

	Criterion	Evidential basis
C1	Task authenticity	Experiential learning and simulation transfer (§2.1)
C2	Candour of occupational representation	Realistic job preview evidence (§2.1); decision-difficulty taxonomy (§1.1)
C3	Adaptivity to the learner	Adaptive instruction and the generative literature (§2.5)
C4	Validity of competence measurement	Evidence-centered design (§2.2)
C5	Quality of feedback and guidance	Assessment is inert without actionable response (follows from C4)
C6	Sustained engagement	Gamification and self-determination literatures (§2.3)
C7	Support for pathway navigation	Career theory and competency-based progression (§2.2, §2.4)
C8	Scalability and currency of content	The authoring literature (§2.5)

A ninth attribute — existence of empirical validation — is discussed rather than tabulated, since it concerns evidence for a system rather than a property of it.

The comparison, and the gap

Criterion	Authored simulation (Forge)	Generated simulation (EntryLevel, Anthropos)	Interview practice (careersim.ai)	Generative narrative (Hidden Door)	Assessment & orientation (O*NET, VN market)
C1 Task authenticity	●	●	◐	○	○
C2 Candour of representation	●	◐	○	○	◐
C3 Adaptivity to the learner	○	●	●	●	◐
C4 Validity of measurement	○	◐	◐	○	○
C5 Feedback and guidance	◐	◐	●	○	◐
C6 Sustained engagement	○	○	○	●	○
C7 Pathway navigation	○	○	○	◐	●
C8 Scalability and currency	○	●	●	●	◐

● Full — a core designed property ◐ Partial — limited, incidental, or a by-product ○ Absent · Table 3.1 also carries a JobQuest column, marked as design intent. Per-cell annotations: Appendix A1.

The gap. Not a missing feature but an unrealised combination: performed, candid occupational tasks that adapt to the individual, from which competence is measured by validated automated judgment, connected by a structure relating occupations through shared competencies, and maintained through expert authoring.

What JobQuest is, and who it serves

Occupations as performable tasks; competence assessed from the performance.

Three presentation conventions— each chosen against evidence

RPG progression

Levels earned by demonstrated competence. Chosen because role-play shows the strongest genre-level association with skill acquisition — overall effect ≈ 0.82 , largest component on skill acquisition. [26]

Visual-novel branching scenes

Narrative is among the strongest moderators in the gamification meta-analyses.

Roguelike career and life events

Narrative variety only — explicitly not coupled to any reward schedule or monetisation, per the harm evidence on randomised rewards. [30]

Four user groups

Explorer

The end user. Everything else exists to serve them.

Domain expert

Authors seeds and rubrics, validates generated scenarios — the guarantor of our central claim.

Content administrator

Catalogue lifecycle, coverage, currency.

System administrator

Model configuration, capacity, cost, data protection.

[26] Role-play methods: overall effect ≈ 0.82 , largest component on skill acquisition. [30] Harm evidence on randomised reward mechanisms.

One mission, end to end

We author the seed. The model expands it. Task types are assessed differently.

- 1 Role and context** Backend Engineer, starter level. Your team, your stack — and tonight you are on call.
- 2 Seed (we author)** The incident skeleton, the required facts, the correct-handling rubric.
- 3 AI expands** Specific symptoms, narrative and tone. Varied each run.
- 4 Choice point** “Payments are down at 2 a.m. — where do you look first?” Branches the story.
- 5 Structured task** Order the triage steps. Scored deterministically against a key.
- 6 Free-response task** Write the incident summary for your team. Scored against the expert's rubric.
- 7 Result** Score, response-specific feedback, competencies evidenced (debugging, communication), and credit toward the next level.

Only the free-response step is a research risk. That is what research problem RP-2 is about.

Nine business rules, and where each comes from

Several are not product decisions but Chapter 2 findings made operational.

Band	Rule	What it requires	Where it comes from
Progression	BR-01	Every occupation's entry level is accessible without prerequisite	<i>Exploration is the purpose; gating discovery defeats it</i>
	BR-02	Higher levels require the specified competence credit	<i>Competency-based progression (§2.2)</i>
	BR-03	Entry to an adjacent occupation requires demonstrated competencies overlapping its entry requirements	<i>Occupational taxonomy (§2.2, §2.6)</i>
Integrity	BR-04	Only a user who has completed an occupation may rate it	<i>Rating validity</i>
	BR-05	No seed or rubric is published without domain-expert validation	<i>Authoring literature (§2.5)</i>
	BR-09	The automated assessor must be drawn from a different model family than the generator	<i>Self-preference bias (§2.5)</i>
Ethics	BR-06	Assessment is formative: never transmitted to employers, never determining access to any external opportunity	<i>Responsibility considerations (§2.7)</i>
	BR-07	Occupational representation must include the unfavourable aspects of the work	<i>Realistic job preview evidence (§2.1)</i>
	BR-08	Statistics derived from user performance are published only in aggregate	<i>Responsibility considerations (§2.7)</i>

Five research problems

RP-1 and RP-2 are the two the project stands or falls on.

RP-1

Constrained generation

Given an expert seed of required facts and target competencies, how should generation be constrained so scenarios stay occupationally accurate and keep eliciting the targets?

RP-2

Validity of automated assessment

To what extent does rubric-based automated judgment of free text agree with expert judgment — chance-corrected and stratified by difficulty?

RP-3

Competence-based adjacency

How should demonstrated competencies map onto occupational adjacency so an unlock is defensible rather than arbitrary? Taxonomies supply skill compositions but not readiness thresholds.

RP-4

Engagement, not displacement

Which mechanisms sustain exploration without displacing the task authenticity that learning depends on?

RP-5

Evaluation methodology

How can a reference set be built, at feasible cost in one field, that supports claims about RP-1 and RP-2 without a large-scale user study?

Six difficulties we are not hiding

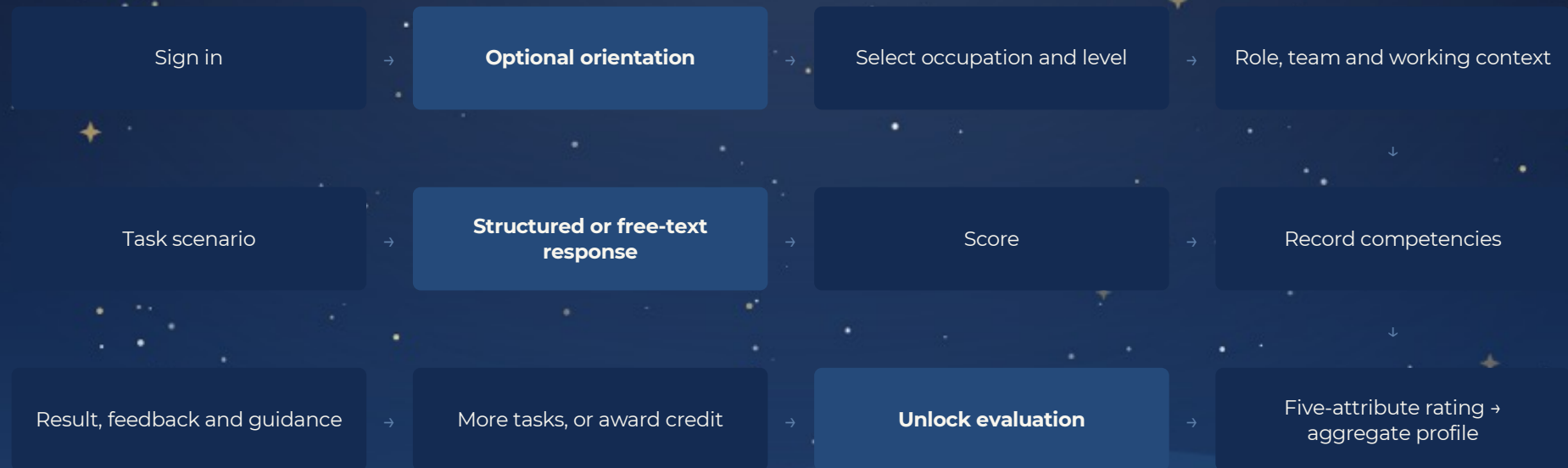
Each one stated plainly, with what we are doing about it.

Difficulty	Why it matters	Our response
Generation fidelity	A plausible but inaccurate scenario is worse than none — it conveys a confident false impression of a real profession. Long-horizon consistency compounds this within multi-task levels.	<i>Bounded authored components; expert validation (BR-05)</i>
Assessment validity	Agreement figures for automated judgment are frequently overstated and degrade unevenly with difficulty.	<i>Stratified expert-authored reference set — expensive to produce</i>
Expert authoring cost	Each occupation requires expert time for seed, rubric and review.	<i>Why phase 1 is one field, and why authoring tooling is a first-class component</i>
Cold start	Aggregate occupational profiles need a population of completions before they are informative.	<i>The interface states the condition rather than displaying an unfounded figure (UC-05 exception flow)</i>
Evaluation without deployment	Offline benchmarking cannot establish engagement retention or effects on career decisions.	<i>Claims bounded accordingly</i>
Localisation	Occupational taxonomies are authored in English for other labour markets and do not map cleanly onto Vietnamese practice.	<i>Expert adaptation, not translation (DR-01)</i>

The two highlighted are the binding ones: expert authoring cost is why phase 1 is a single field, and localisation compounds it.

The exploration process

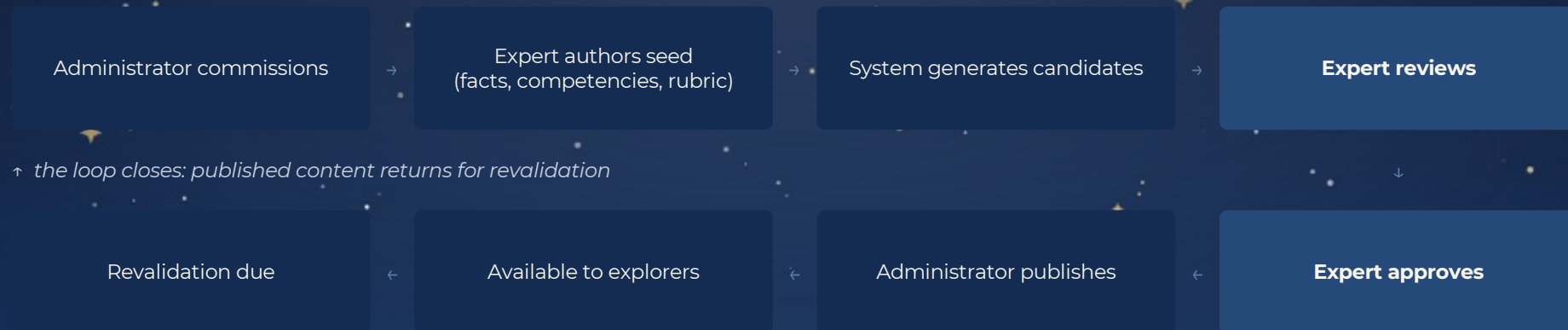
The primary user-facing process, with three decision points.



Navy blocks are the three decision points: the optional orientation (deliberately not a prerequisite — making it one would turn us back into an assessment instrument), the response type (deterministic or rubric-based scoring), and the unlock evaluation (where BR-02 and BR-03 bite).

The content lifecycle

BR-05 made operational — nothing reaches a user without expert validation.



The rejection path branches. An expert may reject a single generated scenario, or correct the seed — in which case every prior generation from that seed is invalidated and regenerated. The seed is where the required facts live.

Where the cost constraint lives. Expert time limits how many occupations we can cover — which is exactly why phase 1 is one field.

Use cases and traceability

Thirteen use cases. Two of them carry the research.

The model

4 actors

Explorer · Domain expert · Content administrator · System administrator

2 external systems

Model provider · Occupational data source

13 use cases

UC-03 Perform a task scenario and UC-08 Validate generated scenario are the two the research problems depend on.

UC-03 exception flow

If generation fails or violates seed constraints, the system retries — then serves a previously validated scenario instance rather than an unvalidated one.

UC-08 exception flow

If the expert judges a scenario passable without the target competency, the seed returns for revision — because the evidence model would otherwise be invalid.

Traceability matrix maps all 43 requirements to the use case realising them — no requirement unrealised, no use case introducing unspecified behaviour. Remaining nine use cases are specified in the same tabular form in Appendix 1.

How we will prove it works

Two constructs, measured separately, against an expert-built reference set — offline.

1

The reference set

Expert-authored scenarios paired with gold-standard judgments, stratified by difficulty, with inter-expert agreement recorded (DR-09). Contains no personal data and is retained as a research asset.

2

What we measure, separately

Generation quality — does produced content preserve the seed's required facts and elicit the intended constructs? Assessed by expert review against specified criteria.

Assessment accuracy — agreement between automated and expert judgment of the same response, reported as Cohen's κ , stratified by item difficulty, alongside test-retest consistency.

3

Why separately

There are three distinct failure modes: biased judgment, unstable judgment, and judgment failing selectively on hard items. They are separately diagnosable only when reported separately — a single accuracy figure hides which one occurred.

What the numbers will realistically look like

Moderate correlations are the realistic expectation, not a disappointment.

The closest methodological precedent

0.28–0.33

correlation between behavioural estimates and external outcomes [56]

≈ 0

correlation against prior attainment — evidence of discriminant validity [56]

The second figure is as informative as the first: it shows the measure is not simply re-measuring prior ability. If we came back with 0.8, I would suspect our reference set rather than celebrate.

What offline evaluation cannot establish

- ✗ Engagement retention over time
- ✗ Effects on actual career decisions
- ✗ Whether simulated experience transfers to workplace performance

Each requires longitudinal study beyond this project's horizon. We would rather state that boundary in the report than have it asked at the defence.

[56] Multi-agent architecture performing unobtrusive assessment in a serious game — the closest methodological precedent.

Where we are, what is next, and the three asks

Chapters 1 through 5.1 are complete and internally consistent.

Delivered

- Evidence base (Ch 2)
- Related work with derived criteria (Ch 3)
- System specification — 9 rules, 43 requirements (Ch 4)
- Process and use case models with traceability (Ch 5.1)

Next

- Complete the Chapter 5 design sections
- Finalise the task-format-to-judging mapping
- Build the expert reference set with difficulty strata
- Implement the authoring interface as a first-class component
- Re-verify the Vietnamese market claim close to submission

The three asks

- 1 **Criteria set and ratings (Ch 3)**
- 2 **Must-priority scope (Ch 4)**
- 3 **Offline evaluation design (Ch 5)**

"Try the job before you pick the career."

APPENDIX A1

Table 3.1 in full — criteria C1 to C4

Criterion	Authored simulation (Forage)	Generated simulation (EntryLevel, Anthropos)	Interview practice (careersim.ai etc.)	Generative narrative (Hidden Door etc.)	Assessment & orientation (O*NET, VN market)
C1 Task authenticity	• employer-authored real work products	• generated role tasks	⦿ interview talk, not work	◦ fictional worlds	◦ inventory only
C2 Candour of representation	• includes unglamorous tasks	⦿ varies by generation	◦ not its purpose	◦ no real occupations	⦿ text lists difficulties
C3 Adaptivity to the learner	◦ fixed content, identical on repeat	• parameterised generation	• adaptive personas	• fully generative	⦿ branching by score
C4 Validity of competence measurement	◦ model answer, no judgment	⦿ behavioural evidence, unvalidated	⦿ scored, conversational only	◦ none	◦ self-report only

- Full — a core designed property.
- ⦿ Partial — limited in scope, incidental, or a by-product of another objective.
- Absent.

APPENDIX A2

Table 3.1 in full — criteria C5 to C8

Criterion	Authored simulation (Forage)	Generated simulation (EntryLevel, Anthropos)	Interview practice (careersim.ai etc.)	Generative narrative (Hidden Door etc.)	Assessment & orientation (O*NET, VN market)
C5 Quality of feedback and guidance	● exemplar, not response-specific	● automated tutor	● per-dimension coaching	○ none	● generic occupation reports
C6 Sustained engagement	○ one-pass certificate	○ discrete projects	○ practice utility	● high replayability	○ single sitting
C7 Support for pathway navigation	○ isolated simulations	○ unconnected projects	○ single-role focus	● world, not career, traversal	● ranked occupational lists
C8 Scalability and currency	○ commissioned per employer	● generated on demand	● generated on demand	● generated on demand	● periodic database updates

- Full — a core designed property.
- Partial — limited in scope, incidental, or a by-product of another objective.
- Absent

APPENDIX A3

Functional requirements (Table 4.2) — FR-01 to FR-11

ID	Requirement	Group	Priority	Feasibility	Verification method
FR-01	Register and authenticate a user account	Explorer	Must	High	Account creation and sign-in succeed; unauthenticated access to progress is refused
FR-02	Present the entry level of any published occupation without prerequisite	Explorer	Must	High	Entry level reachable for a new account with zero credit
FR-03	Present role, team and working context before the first task of a level	Explorer	Must	High	Context screen precedes first task in every level
FR-04	Generate a task scenario from a published seed, preserving its required facts	Explorer	Must	Medium	Expert review of sampled generations against the seed's fact list
FR-05	Accept structured responses and score them against an answer key	Explorer	Must	High	Known-input tests produce expected scores
FR-06	Accept free-text responses and score them against the seed's rubric	Explorer	Must	Medium	Agreement with expert judgment on the reference set (Cohen's κ)
FR-07	Return result, response-specific feedback and forward guidance as distinct elements	Explorer	Must	High	All three present and distinguishable for every assessed response
FR-08	Record competencies evidenced by each assessed response	Explorer	Must	Medium	Competence records correspond to rubric criteria for sampled sessions
FR-09	Award competence credit on level completion and unlock levels per BR-02	Explorer	Must	High	Threshold boundary tests unlock at and only at the specified credit
FR-10	Offer entry to adjacent occupations when overlap conditions are met per BR-03	Explorer	Should	Medium	Offered set matches taxonomy-derived expectation for seeded profiles
FR-11	Conduct an optional orientation conversation producing candidate occupations	Explorer	Should	Medium	Conversation completes and yields a non-empty ranked candidate set

APPENDIX A4

Functional requirements (Table 4.2) — FR-12 to FR-21

ID	Requirement	Group	Priority	Feasibility	Verification method
FR-12	Collect a five-attribute rating on completion and display aggregates	Explorer	Should	High	Rating accepted only after completion; aggregate updates; suppression below threshold
FR-13	Present a private record of attempted occupations, competencies and awards	Explorer	Should	High	Record reflects session history for a seeded account
FR-14	Introduce probabilistic career and life events affecting subsequent exploration	Explorer	Could	Medium	Events occur within configured probability bounds across repeated runs
FR-15	Author and edit a seed with required facts, target competencies and rubric	Expert	Must	High	Seed persists and is retrievable with all three components
FR-16	Review, approve, reject or correct generated scenarios	Expert	Must	High	Unapproved scenarios are not served to explorers
FR-17	Report disagreements between automated and expert judgment	Expert	Should	Medium	Disagreement list matches recomputation over the reference set
FR-18	Manage occupation lifecycle: create, track status, publish, retire	Admin	Must	High	State transitions follow the lifecycle; publication requires completed review
FR-19	Report catalogue coverage and last-validated dates	Admin	Should	High	Report matches stored metadata
FR-20	Configure generation and assessment model providers, enforcing separation	Admin	Must	Medium	Configuration rejects identical provider for both roles
FR-21	Monitor usage, cost and error rates	Admin	Should	High	Metrics present and consistent with generated load

APPENDIX A5

Non-functional requirements (Table 4.3)

ID	Requirement	Category	Priority	Feasibility	Verification method
NFR-01	Scenario generation completes within 5 s at the 95th percentile; progress indicator beyond 2 s	Performance	Must	Medium	Latency distribution over ≥ 200 sampled generations
NFR-02	Assessment of a free-text response completes within 8 s at the 95th percentile	Performance	Must	Medium	Latency distribution over ≥ 200 sampled assessments
NFR-03	Session progress survives interruption and resumes without loss	Reliability	Must	High	Forced interruption tests resume at the recorded position
NFR-04	On model provider failure, no committed progress is lost; user notified within 10 s	Reliability	Should	Medium	Fault injection; zero progress-record loss across injected failures
NFR-05	Personal data collected only as required, stored encrypted, deletable on request	Security, privacy	Must	High	Data inventory review; deletion request removes records
NFR-06	Aggregate statistics apply a minimum-population suppression threshold	Security, privacy	Must	High	Aggregates below threshold are withheld
NFR-07	Machine generation and machine assessment are disclosed to the user	Transparency	Must	High	Disclosure present at the point of use
NFR-08	Interface language and occupational content support Vietnamese and English	Usability	Should	Medium	Both locales render complete content
NFR-09	$\geq 80\%$ of representative readers correctly state a task's objective after reading it	Usability	Should	Medium	Comprehension check with a minimum of 20 readers
NFR-10	Adding an occupation requires no code modification	Maintainability	Must	High	A new occupation is published through the authoring interface alone
NFR-11	Mean model cost per completed session stays within the configured ceiling	Cost	Should	Medium	Cost telemetry aggregated per session against the ceiling
NFR-12	Model provider is replaceable without changes to seeds or rubrics	Portability	Should	Medium	Provider substitution test with unchanged content

APPENDIX A6

Data requirements (Table 4.4)

ID	Data requirement	Source	Priority	Feasibility	Verification method
DR-01	Occupational reference data: occupations, tasks, work activities and skills	O*NET and ESCO, adapted for Vietnamese practice by experts	Must	Medium	Imported records resolve against source identifiers
DR-02	Competency taxonomy with definitions and observable indicators	Derived from DR-01 under expert adaptation	Must	Medium	Every rubric criterion resolves to a taxonomy entry
DR-03	Occupational adjacency relations with overlap measures	Computed from DR-01 / DR-02	Should	Medium	Computed adjacency reviewed by experts for plausibility
DR-04	Scenario seeds: required facts, target competencies, narrative skeleton, rubric	Domain experts	Must	High	Seed schema validation; expert sign-off recorded
DR-05	Generated scenario instances with provenance to seed and model version	System	Must	High	Every served scenario carries resolvable provenance
DR-06	User profile and progress: credit, unlocked levels, occupation history	System	Must	High	Progress reconstructs from stored records
DR-07	Assessment records: response, score, competencies, feedback, guidance	System	Must	High	Records complete for every assessed response
DR-08	Occupation ratings on five attributes, completion precondition enforced	Explorers	Should	High	Ratings exist only for completed occupations
DR-09	Expert reference set: gold-standard judgments, stratified by difficulty	Domain experts	Must	Medium	Set covers defined strata with recorded inter-expert agreement
DR-10	Operational telemetry: latency, cost, error rates, model versions	System	Should	High	Telemetry present and queryable

Personal data within DR-06, DR-07 and DR-08 is subject to the policies in §4.1.4. DR-09 contains no personal data and is retained as a research asset.

Use cases — the exception flows

Full specifications: Tables 5.1–5.5; the remaining nine in Appendix 1.

1

UC-03 Perform a task scenario

Exception flow — if generation fails or violates seed constraints, the system retries, then serves a previously validated scenario instance rather than an unvalidated one.

Requirements: FR-03 to FR-08 · NFR-01 to NFR-04, NFR-07 · DR-04, DR-05, DR-07

2

UC-08 Validate generated scenario

Exception flow — if the expert judges a scenario passable without the target competency, the seed returns for revision, because the evidence model would otherwise be invalid.

Requirements: FR-16 · NFR-10 · DR-04, DR-05. Business rule: BR-05

3

UC-05 Rate a completed occupation

Precondition: the user has completed the occupation (BR-04).

Exception flow — population below the suppression threshold: the aggregate is withheld and the condition is stated rather than displaying an unfounded figure.

Requirements: FR-12 · NFR-06 · DR-08. Business rules: BR-04, BR-08

APPENDIX A8

Traceability: requirements to use cases (Table 5.6)

Use case	Functional	Non-functional	Data
UC-01 Explore catalogue	FR-02	NFR-08, NFR-09	DR-01
UC-02 Orientation conversation	FR-11	NFR-01, NFR-07	DR-01, DR-06
UC-03 Perform task scenario	FR-03, FR-04, FR-05, FR-06, FR-07, FR-08	NFR-01, NFR-02, NFR-03, NFR-04, NFR-07	DR-04, DR-05, DR-07
UC-04 Progress further	FR-09, FR-10, FR-14	—	DR-02, DR-03, DR-06
UC-05 Rate occupation	FR-12	NFR-06	DR-08
UC-06 Review personal record	FR-13	NFR-05	DR-06, DR-07
UC-07 Author seed and rubric	FR-15	NFR-10	DR-01, DR-02, DR-04
UC-08 Validate scenario	FR-16	NFR-10	DR-04, DR-05
UC-09 Review disagreements	FR-17	—	DR-07, DR-09
UC-10 Manage lifecycle	FR-18	NFR-10	DR-04
UC-11 Monitor coverage	FR-19	—	DR-04
UC-12 Configure models	FR-20	NFR-11, NFR-12	DR-05
UC-13 Monitor operation	FR-21	NFR-11	DR-10
Account management	FR-01	NFR-05	DR-06

Every requirement is realised somewhere; no use case introduces behaviour the requirements do not specify.

APPENDIX A9

References cited in this deck

Numbering matches the report's bibliography. [1]–[5] are given in full.

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[4] T. Q. Tran, N. B. T. Vu, D. Van Le and L. H. Vu, "Vertical and horizontal job-education mismatches and wages," *International Journal of Educational Development*, vol. 117, art. 103323, 2025.

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[17][18] Evidence-centered design — Mislevy et al.

[24] Li, Hew & Du — gamification by psychological need · [26] role-play methods · [27] Sitzmann

[30] Randomised reward mechanisms and problem gambling

[32] Holland · [33] Super · [34] Krumboltz

[44] Zheng et al. (MT-Bench) · [45] rubric-anchored chain-of-thought · [46] 21 evaluators, ~541,000 judgments · [47] JudgeBench · [51] RewardBench

[49] O*NET · [50] ESCO · [52] EQGBench · [53] EduBench · [54] TextWorld · [55] Jericho

[56] Multi-agent unobtrusive assessment in a serious game

[57] CareerSim · [58] Future You · [59] Letters from Future Self · [60] CareerPooler · [63] SDT-grounded counselling agent

[64] Forage · [65][66] EntryLevel, Anthropos · [70] Hidden Door

[71–75] JobWay, JobTest.vn, Hướng nghiệp Sông An, Hướng Nghiệp LwL, AI Hay

APPENDIX A10

List of abbreviations

DACN Đồ án chuyên ngành — Specialized Project

BR Business rule

UC Use case

ECD Evidence-centered design

SCCT Social cognitive career theory

LLM Large language model

ESCO European Skills, Competences, Qualifications and Occupations

κ (kappa) Cohen's kappa — chance-corrected agreement

HCMUT Ho Chi Minh City University of Technology

FR / NFR / DR Functional / non-functional / data requirement

RP Research problem

RJP Realistic job preview

SDT Self-determination theory

O*NET Occupational Information Network (US Dept. of Labor)

MBTI Myers–Briggs Type Indicator

g Hedges' g — standardised mean difference effect size